RISK MANAGEMENT PLAN

Kelvaparib (KLV-3308)

Marnholt Oncology GmbH

SCOPE OF THIS DOCUMENT

This document covers the complete clinical programme. The studies within scope, and the number of subjects contributing data to each, are listed below.

Table 0-1 Studies Within the Scope of This Document

Study	Phase	Design	Subjects treated
KLV-7704	Phase 2	single arm	81
KLV-7701	Phase 3	randomised	627
KLV-7702	Phase 1	escalation	58
KLV-7703	Phase 3	randomised	354
Total			1120

Subject counts are those treated with at least one dose within each study.

Studies in the clinical programme

Table 1 Overview of Clinical Studies in the KLV-3308 Program

Study	Phase	Design	Population	Subjects enrolled	Treatment	Primary endpoint	Status
KLV-7704	Phase 2	Single-arm, open-label	recurrent HGSOC, platinum-sensitive or platinum-resistant	81	Kelvaparib 300 mg	Objective response rate by investigator assessment	Completed
KLV-7701	Phase 3	Randomised, controlled, parallel group	recurrent ovarian carcinoma after platinum-based therapy	627	Kelvaparib 300 mg vs Placebo	Progression-free survival by investigator assessment	Completed
KLV-7702	Phase 1	Open-label dose escalation	advanced solid tumours with a homologous recombination repair defect	58	100 mg, 200 mg,300 mg, 400 mg	Maximum tolerated dose and recommended phase 2 dose	Completed
KLV-7703	Phase 3	Randomised, controlled, parallel group	recurrent ovarian carcinoma, platinum-sensitive or platinum-resistant	354	Kelvaparib 300 mg vs Placebo	Progression-free survival by investigator assessment	Completed

Subject counts are the number enrolled and treated in each study. Study status is as at the date of this document.

1. SAFETY SPECIFICATION

Important identified risks are Myelosuppression, myelodysplastic syndrome or acute myeloid leukaemia, and pneumonitis. Important potential risks are second primary malignancy and effects on fertility. Missing information includes use in severe hepatic impairment, use in pregnancy and long-term safety beyond three years of exposure.

2. PHARMACOVIGILANCE PLAN

Routine pharmacovigilance applies to all identified and potential risks. Additional activities comprise a dedicated hepatic impairment study and a long-term follow-up registry.

3. RISK MINIMISATION MEASURES

Routine measures comprise the product information, including a full blood count before the first dose, weekly for the first month and monthly thereafter, and the rule to withhold treatment for a grade 3 or higher cytopenia and resume at a reduced dose (section 4.4 of the product information). Additional measures comprise a prescriber guide and a patient card describing the monitoring schedule.

4. SUMMARY

The plan is proportionate to the identified risks and relies principally on the monitoring and dose-modification measures validated during clinical development.

5 PART I: PRODUCT OVERVIEW

Kelvaparib (KLV-3308), an oral PARP1-selective inhibitor, for high-grade serous ovarian carcinoma. The formulation, route and schedule are as described in the protocols and in the proposed labelling.

This plan covers the product across the indications under development. It is revised whenever new information changes the characterisation of a risk, whenever a risk minimisation measure changes, and at the intervals required by the applicable regulation.

6 PART II MODULE SI: EPIDEMIOLOGY OF THE INDICATION

The indication, the population affected, the usual course of the disease and the outcomes achieved with current management are described. Estimates vary with how the population is selected and how long it is followed, and that variability is stated rather than concealed by quoting a single figure.

Important comorbidities in the population are described, because they determine which events will occur in the population irrespective of treatment and therefore which events require careful attribution.

7 PART II MODULE SII: NON-CLINICAL SAFETY SPECIFICATION

The non-clinical findings relevant to human safety are summarised: target organ toxicity, safety pharmacology, genotoxicity, and the status of the reproductive, developmental and carcinogenicity work. Findings that are monitorable in the clinic are distinguished from those that are not.

8 PART II MODULE SIII: CLINICAL TRIAL EXPOSURE

Exposure is presented by study, by dose, by duration and by demographic characteristic, so that the size of the population in which each risk has been observed can be judged. Exposure in subjects receiving comparator or placebo is presented separately.

9 PART II MODULE SIV: POPULATIONS NOT STUDIED

The populations excluded by the eligibility criteria are listed with the reason for each exclusion and the consequence for what is known: children, pregnant and breastfeeding women, subjects with significant organ impairment, and subjects with the conditions excluded on safety grounds. Absence of data is recorded as missing information rather than as evidence of safety.

10 PART II MODULE SV: POST-AUTHORISATION EXPERIENCE

The product is under development and there is no post-authorisation experience. This module is completed when that changes.

11 PART II MODULE SVI: ADDITIONAL REQUIREMENTS

Any requirement arising from the applicable regulation that is not covered elsewhere in this plan is addressed here, including potential for misuse and for off-label use.

12 PART II MODULE SVII: IDENTIFIED AND POTENTIAL RISKS

The risks below are those for which evidence exists or for which the mechanism gives reason to expect them. Each is stated with the evidence, the affected population, the usual presentation and the measures in place.

Important identified risk: Myelodysplastic syndrome or acute myeloid leukaemia

Evidence. Myelodysplastic syndrome or acute myeloid leukaemia was reported in 13 of the 793 kelvaparib-treated subjects across the programme (1.6 per cent): grade 3 or higher in 8, serious in 7 and fatal in 1, with a median time to first onset of 29 days (study reports, Section 12.2.6). Mechanism. Consistent with the pharmacology of an oral PARP1-selective inhibitor. Population. All treated subjects, with higher risk in those identified by the protocol criteria. Presentation. As described in the protocol management guidance. Measures. Eligibility restriction, scheduled monitoring, mandatory dose modification, management guidance and expedited reporting.

Important identified risk: Pneumonitis

Evidence. Pneumonitis was reported in 29 of the 793 kelvaparib-treated subjects across the programme (3.7 per cent): grade 3 or higher in 11, serious in 10 and fatal in 2, with a median time to first onset of 42 days (study reports, Section 12.2.6). Mechanism. Consistent with the pharmacology of an oral PARP1-selective inhibitor. Population. All treated subjects, with higher risk in those identified by the protocol criteria. Presentation. As described in the protocol management guidance. Measures. Eligibility restriction, scheduled monitoring, mandatory dose modification, management guidance and expedited reporting.

Important identified risk: Severe anaemia

Evidence. Severe anaemia was reported in 130 of the 793 kelvaparib-treated subjects across the programme (16.4 per cent): grade 3 or higher in 130, serious in 76 and fatal in 8, with a median time to first onset of 38 days (study reports, Section 12.2.6). Mechanism. Consistent with the pharmacology of an oral PARP1-selective inhibitor. Population. All treated subjects, with higher risk in those identified by the protocol criteria. Presentation. As described in the protocol management guidance. Measures. Eligibility restriction, scheduled monitoring, mandatory dose modification, management guidance and expedited reporting.

13 PART II MODULE SVIII: MISSING INFORMATION

Use in pregnancy and lactation; use in children; use in subjects with significant renal or hepatic impairment; long-term exposure beyond the duration studied. Each is missing information rather than a risk, and each is addressed by the plan below.

14 PART III: PHARMACOVIGILANCE PLAN

Routine pharmacovigilance applies to all risks: collection, assessment and expedited reporting of serious adverse reactions, periodic aggregate review, and signal detection against the cumulative database.

Additional pharmacovigilance activity is proposed where routine activity would not answer the question: targeted follow-up questionnaires for the events of special interest, and the studies listed in the development plan that address the missing information above.

15 PART IV: POST-AUTHORISATION EFFICACY STUDIES

None is proposed at this stage of development. This part is completed if a commitment arises during the authorisation procedure.

16 PART V: RISK MINIMISATION MEASURES

Routine risk minimisation is by the product information: the contraindications, warnings, monitoring instructions and dose modification advice that follow from the risks above.

Additional risk minimisation is proposed only where routine measures would be insufficient. Where proposed, the measure, its objective and the way its effectiveness will be measured are stated together, because a measure whose effectiveness cannot be measured cannot be evaluated.

17 PART VI: SUMMARY FOR THE PUBLIC

A summary of the plan in plain language is prepared for publication, covering what the medicine is for, the important risks, what is not yet known, and what is being done to find out.

18 PART VII: ANNEXES

The annexes hold the protocols of the studies in the pharmacovigilance plan, the specific adverse reaction follow-up forms, and the details of the proposed risk minimisation measures.

19 MISSING INFORMATION IN DETAIL

Use in pregnancy and lactation. Excluded by the eligibility criteria, and the reproductive and developmental package is incomplete. Addressed by contraception requirements, by pregnancy reporting and follow-up to outcome, and by the studies planned in the development plan.

The activity above is recorded with the risk or the gap it addresses, the milestone against which progress is judged, and the point at which it will be reviewed. An activity without those three is an intention rather than a plan.

Use in children. Not studied. Addressed by the paediatric investigation plan agreed with the authorities, which sets out when and how a paediatric population will be studied.

The activity above is recorded with the risk or the gap it addresses, the milestone against which progress is judged, and the point at which it will be reviewed. An activity without those three is an intention rather than a plan.

Use in significant renal or hepatic impairment. Subjects with significant impairment were excluded. Exposure in impairment is addressed by the population pharmacokinetic analysis and, where that is insufficient, by a dedicated study in the development plan.

The activity above is recorded with the risk or the gap it addresses, the milestone against which progress is judged, and the point at which it will be reviewed. An activity without those three is an intention rather than a plan.

Long-term exposure. Exposure beyond the duration studied is unknown. Addressed by continued follow-up of subjects who remain on treatment and by the periodic aggregate review of the cumulative safety database.

The activity above is recorded with the risk or the gap it addresses, the milestone against which progress is judged, and the point at which it will be reviewed. An activity without those three is an intention rather than a plan.

20 PHARMACOVIGILANCE ACTIVITIES IN DETAIL

Routine pharmacovigilance. Collection, assessment, coding and expedited reporting of serious adverse reactions; periodic aggregate review; signal detection against the cumulative database; and maintenance of the reference safety information.

The activity above is recorded with the risk or the gap it addresses, the milestone against which progress is judged, and the point at which it will be reviewed. An activity without those three is an intention rather than a plan.

Targeted follow-up questionnaires. Applied to each event of special interest so that the information needed to characterise the event is collected consistently rather than left to the free text of the initial report.

The activity above is recorded with the risk or the gap it addresses, the milestone against which progress is judged, and the point at which it will be reviewed. An activity without those three is an intention rather than a plan.

Cumulative review at defined intervals. Each identified and potential risk is reviewed against the cumulative data at each periodic report, and the characterisation is revised where the data warrant it.

The activity above is recorded with the risk or the gap it addresses, the milestone against which progress is judged, and the point at which it will be reviewed. An activity without those three is an intention rather than a plan.

Studies addressing missing information. The studies in the development plan that address the gaps listed above, each with its objective, milestones and the question it is intended to answer.

The activity above is recorded with the risk or the gap it addresses, the milestone against which progress is judged, and the point at which it will be reviewed. An activity without those three is an intention rather than a plan.

21 RISK MINIMISATION MEASURES IN DETAIL

Product information. The contraindications, warnings, monitoring instructions and dose modification advice that follow from the identified risks. This is the primary risk minimisation measure and applies to every risk.

The activity above is recorded with the risk or the gap it addresses, the milestone against which progress is judged, and the point at which it will be reviewed. An activity without those three is an intention rather than a plan.

Prescriber guidance. Where routine product information would be insufficient, guidance directed at the prescriber describing the monitoring required and the action to take. Proposed only where justified.

The activity above is recorded with the risk or the gap it addresses, the milestone against which progress is judged, and the point at which it will be reviewed. An activity without those three is an intention rather than a plan.

Patient information. Material directed at the subject describing what to watch for and when to seek advice, in language tested for readability.

The activity above is recorded with the risk or the gap it addresses, the milestone against which progress is judged, and the point at which it will be reviewed. An activity without those three is an intention rather than a plan.

Measurement of effectiveness. Every additional measure is proposed together with how its effectiveness will be measured. A measure whose effectiveness cannot be measured cannot be evaluated and is not proposed.

The activity above is recorded with the risk or the gap it addresses, the milestone against which progress is judged, and the point at which it will be reviewed. An activity without those three is an intention rather than a plan.